

DNS AND DHCP

1. Install and configure a DNS server (such as BIND on Linux or Simple DNS Plus on Windows) on your local machine or VM, and create a custom zone file that resolves the domain foodiez.in to the IP address 192.168.1.50.

-> **Install and configure DNS with BIND9**

Step 1: Install BIND9

```
sudo apt update
```

```
sudo apt install bind9 bind9-utils dnsutils
```

Check that BIND is running:

```
sudo systemctl status bind9
```

Step 2: Create the DNS zone

Edit:

```
sudo nano /etc/bind/named.conf.local
```

Add:

```
zone "foodiez.in" {  
    type master;  
    file "/etc/bind/db.foodiez.in";  
};
```

Save the file.

Step 3: Create the zone file

Create:

```
sudo nano /etc/bind/db.foodiez.in
```

Enter:

```
$TTL 604800
```

```
@    IN    SOA    ns1.foodiez.in. admin.foodiez.in. (  
                2026092901
```

604800
86400
2419200
604800)

@ IN NS ns1.foodiez.in.

ns1 IN A 192.168.1.50

@ IN A 192.168.1.50

The important record is:

@ IN A 192.168.1.50

This means:

foodiez.in → 192.168.1.50

Step 4: Check the configuration

Run:

```
sudo named-checkconf
```

Then:

```
sudo named-checkzone foodiez.in /etc/bind/db.foodiez.in
```

You should see something similar to:

```
zone foodiez.in/IN: loaded serial 2026092901
```

OK

Restart BIND:

```
sudo systemctl restart bind9
```

2. Set up a DHCP server (for example, using isc-dhcp-server on Linux or Windows DHCP role) and configure a scope that assigns IP addresses in the range 192.168.1.100 to 192.168.1.150 with a lease time of 2 hours.

-> Install ISC DHCP:

```
sudo apt install isc-dhcp-server
```

Edit the DHCP configuration:

```
sudo nano /etc/dhcp/dhcpd.conf
```

Add:

```
default-lease-time 7200;
```

```
max-lease-time 7200;
```

```
authoritative;
```

```
subnet 192.168.1.0 netmask 255.255.255.0 {  
    range 192.168.1.100 192.168.1.150;  
    option subnet-mask 255.255.255.0;  
    option routers 192.168.1.1;  
    option domain-name-servers 192.168.1.50;  
    option domain-name "foodiez.in";  
}
```

Lease time

The value:

7200

is seconds.

7200 seconds = 2 hours.

Therefore, clients receive a **2-hour DHCP lease**.

Check DHCP configuration

```
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
```

If there are no errors, restart DHCP:

```
sudo systemctl restart isc-dhcp-server
```

Check:

```
sudo systemctl status isc-dhcp-server
```

3. Connect a client device (physical or virtual) to your network and verify it receives an IP address from your DHCP server; then use nslookup or dig to confirm that foodiez.in resolves to the correct IP address from your DNS server

-> Connect another VM to the same isolated virtual network.

On the client, request an IP address.

Linux client

```
sudo dhclient -r
```

```
sudo dhclient
```

Then:

```
ip addr
```

You should receive an address such as:

```
192.168.1.100
```

or another address between:

```
192.168.1.100 – 192.168.1.150
```

Check the DHCP lease:

```
ip route
```

and:

```
cat /var/lib/dhcp/dhclient.leases
```

Test DNS

Use nslookup:

```
nslookup foodiez.in 192.168.1.50
```

Expected result:

```
Server:    192.168.1.50
```

```
Address:    192.168.1.50#53
```

```
Name:      foodiez.in
```

```
Address:    192.168.1.50
```

You can also use:

```
dig @192.168.1.50 foodiez.in
```

Look for:

ANSWER SECTION:

```
foodiez.in. ... IN A 192.168.1.50
```

This confirms:

DHCP → client receives IP

and

DNS → foodiez.in resolves to 192.168.1.50

4. Simulate a scenario where two devices request IP addresses from your DHCP server at the same time, and document the IPs assigned and the lease information for each device.

 Hint: Use the DHCP server's lease file or management console to view active leases and their expiration times.

-> Connect **Client 1** and **Client 2** to the same lab network.

On Client 1:

```
sudo dhclient -r
```

```
sudo dhclient
```

On Client 2:

```
sudo dhclient -r
```

```
sudo dhclient
```

For example, you might get:

Client	Assigned IP	Lease
Client 1	192.168.1.100	2 hours
Client 2	192.168.1.101	2 hours

The exact addresses aren't guaranteed; DHCP chooses available addresses from the configured pool.

On the DHCP server, inspect:

```
sudo cat /var/lib/dhcp/dhcpd.leases
```

You should see lease records resembling:

```
lease 192.168.1.100 {  
    starts 3 2026/09/29 11:30:00;  
    ends 3 2026/09/29 13:30:00;  
    binding state active;  
    hardware ethernet AA:BB:CC:DD:EE:01;  
}
```

```
lease 192.168.1.101 {  
    starts 3 2026/09/29 11:30:02;  
    ends 3 2026/09/29 13:30:02;  
    binding state active;  
    hardware ethernet AA:BB:CC:DD:EE:02;  
}
```

The MAC addresses and exact timestamps will depend on your devices.

What to document

Record:

- Client hostname
- MAC address
- Assigned IP
- Lease start time
- Lease expiration time
- Lease duration

5. Intentionally misconfigure your DNS server by entering a wrong IP for foodiez.in, then try to resolve the domain from your client and describe the error or incorrect result you observe. Finally, fix the error and verify that the resolution works correctly.

-> Intentionally break and fix DNS

Initially, the correct record is:

foodiez.in → 192.168.1.50

Step 1: Introduce an incorrect IP

Edit:

```
sudo nano /etc/bind/db.foodiez.in
```

Change:

```
@    IN    A    192.168.1.50
```

to:

```
@    IN    A    192.168.1.99
```

Increase the serial number:

```
2026092901
```

to:

```
2026092902
```

This is important because DNS uses the serial number to identify a newer zone version.

Check the zone:

```
sudo named-checkzone foodiez.in /etc/bind/db.foodiez.in
```

Then reload BIND:

```
sudo rndc reload foodiez.in
```

or:

```
sudo systemctl restart bind9
```

Step 2: Test the incorrect result

From the client:

```
nslookup foodiez.in 192.168.1.50
```

You should now see:

```
Name:  foodiez.in
```

```
Address: 192.168.1.99
```

This demonstrates that DNS itself is working, but the **DNS record contains the wrong IP address**.

If you instead receive the old answer, DNS caching may be involved. You can wait for the TTL to expire or query the authoritative server directly.

Step 3: Fix the DNS record

Change:

@ IN A 192.168.1.99

back to:

@ IN A 192.168.1.50

Increase the serial again:

2026092903

Check and reload:

sudo named-checkzone foodiez.in /etc/bind/db.foodiez.in

sudo rndc reload foodiez.in

Finally:

nslookup foodiez.in 192.168.1.50

Expected:

Name: foodiez.in

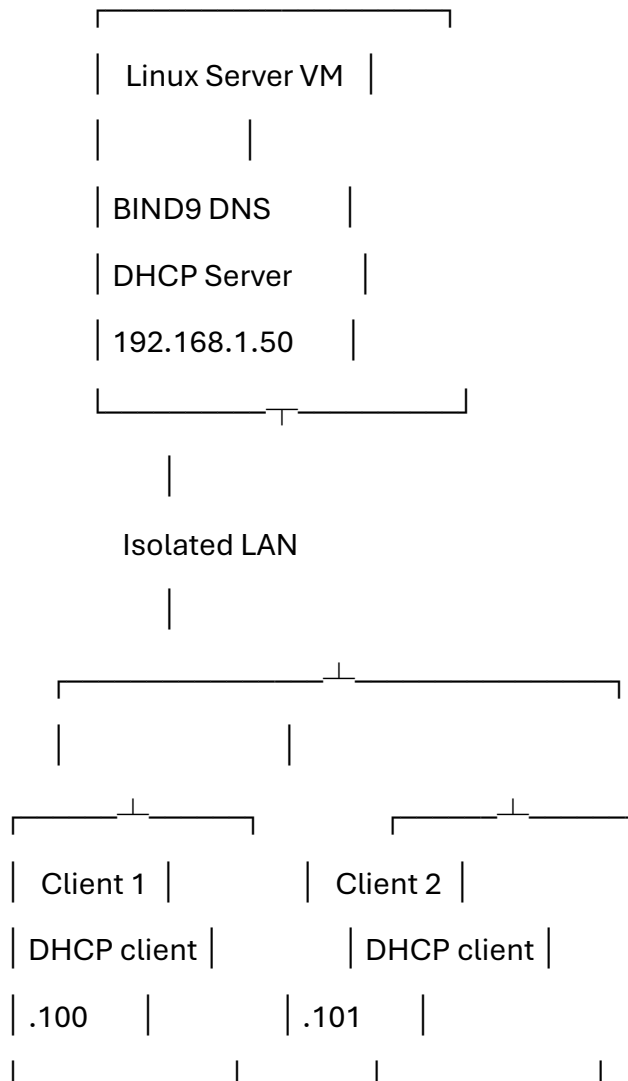
Address: 192.168.1.50

Final Lab Results

Task	Expected Result
DNS configuration	foodiez.in → 192.168.1.50
DHCP scope	192.168.1.100 – 192.168.1.150
DHCP lease	2 hours / 7200 seconds
Client DHCP test	Client receives an address from the scope
DNS test	foodiez.in resolves to 192.168.1.50
Two-client test	Two unique addresses are leased
DNS error test	Incorrect IP is returned after changing the A record

Task	Expected Result
DNS repair	foodiez.in again resolves to 192.168.1.50

Lab topology:



foodiez.in —————▶ 192.168.1.50

This gives you a complete **DNS + DHCP + client verification + troubleshooting** practical lab.